

Role of artificial intelligence in veterinary anatomical diagnostics and zoonotic disease monitoring

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ABSTRACT

Background: The use of artificial intelligence (AI) also playing a significant role in veterinary medicine due to the changing pattern of diseases in terms of climatic changes and advances in treatment protocols. About 60 % of emerging human diseases are zoonotic mainly originating from animals, so the conventional diagnostic tools and traceability protocols are not fast enough, precise and lack of ability to handle large number of cases. Use of AI tools can make a big difference in diagnosis of diseases/problems through diagnostic images, predicting outbreaks through the data from previous records ultimately leading to improved monitoring of zoonotic diseases in terms of early warning systems for future outbreaks, multisectoral collaborations to improve the health of humans, animals and environments. In diagnosis, AI shows great effectiveness, like being able to spot more than 90 % of bone and joint issues in X-rays, predicting sickness in farm animals up to two to three days before they show symptoms, and even predicting animal diseases those can be transmitted to humans up to weeks in advance by looking at data about the environment changes and animals' movements. But adoption to these AI systems is still not common because of many reasons including scattered data, lack of understanding about algorithms, ethical issues, and unequal access to technology etc.

Conclusions: As climate change speeds up the spread of diseases from animals to humans, AI is becoming a crucial tool for reaching health goals that affect both people and animals. But this will happen only if AI is used fairly and responsibly. This summary shows that working together across different fields is important to combine new technology with expert knowledge from vets. The goal is to use AI to support, not take over, what doctors do, and to make advanced care available to everyone around the world.

1. Introduction

In a world where roughly 60 % of emerging human infectious diseases originate from animals (Taylor et al., 2001), the shortcomings of traditional veterinary diagnostic techniques, especially in interpreting complex anatomical variations, have become increasingly evident. Veterinarians today are confronted with a challenging environment marked by rising zoonotic threats, heavy workloads, and the persistent difficulty of diagnosing conditions stemming from subtle anatomical changes. Artificial intelligence (AI) is emerging not as a substitute for veterinary expertise but as a robust collaborator that enhances our

ability to understand the intricate details of animal anatomy. The creation and testing of these AI tools have greatly improved because of animal anatomical teaching models. These models offer important, consistent ways to train and improve diagnostic algorithms (Choudhary and Sarkar, 2025). For instance, AI-assisted interpretation of radiographs can reduce diagnostic errors by up to 30 % in small animal practice, demonstrating superhuman precision (Lakhani and Sundaram, 2017; Zhang et al., 2024). More significantly, recent studies confirm that AI's potential in animal anatomy goes far beyond basic detection, offering a new paradigm for understanding morphological structures and their pathological changes (Choudhary et al., 2025).

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The consequences of wrong diagnosis in anatomy are especially serious when looking at the whole population. The 2018 African swine fever outbreak in Asia, which caused economic losses over \$20 billion, showed how traditional ways of monitoring diseases fail to stop fast-spreading illnesses (Food and Agriculture Organization of the United Nations FAO, 2020). Now, AI systems are changing things a lot; machine learning models use satellite images, climate information, and farm data to predict disease outbreaks up to weeks before they happen. These AI tools in anatomical imaging are changing how we do diagnosis and disease tracking, connecting individual animal health with global public health (Vickram et al., 2025). Still, not many people are using these AI tools. Less than 15 % of veterinary clinics use them, mainly because of high costs, the difficulty in understanding complex algorithms, and the need for better connection with basic knowledge of anatomy.

This review article will provide the information about how AI is changing veterinary medicine in two broader perspectives. First, it helps doctors to understand medical images and records better, making diagnosis more accurate. Second, it's being used to build smart systems those can track zoonotic diseases transmit between animals and humans. It will address the role of technology itself, but also the real challenges when trying to use it, like ethical issues and gap between those who have access to new techniques and those who don't. Due to climate changes emergence of zoonotic diseases and their spread is many folds quicker as compared to other historical outbreaks (Carlson et al., 2022). Hence using AI carefully and smartly could be the ways to make the One Health idea real time implementation. This would help to make this planet safer for the animals, people, and the environment, we all live in. In this article the available data on AI will be reviewed in following sub-headings: (1) AI in Veterinary Anatomical Diagnostics, (2) AI in Zoonotic diseases monitoring, (3) Challenges and (4) Future directions.

2. AI in veterinary anatomical diagnostics

2.1. AI in anatomical imaging

Artificial intelligence is fundamentally transforming veterinary diagnostic imaging through powerful, data-driven insight into animal anatomy. These systems are particularly proficient in identifying subtle pathological changes in anatomical structures from micro-fractures in cortical bone to diffuse parenchymal patterns in internal organs that even may go unnoticed by experienced professionals (Liu et al., 2019). The integration of AI-driven technologies in anatomical imaging is proving to be a game-changer, significantly improving the precision and efficiency of veterinary diagnostics and surgical planning (Vickram et al., 2025).

Deep learning models, especially convolutional neural networks (CNNs), exhibit exceptional performance when trained on large datasets of canine and feline radiographs, accurately detecting fractures, tumors, and osteoarthritic changes with sensitivities often surpassing 90 % (Rossi et al., 2023). In equine medicine, the use of systems like SignalPET, which combine kinematic data with imaging analysis, enables the identification of subtle lameness by linking gait abnormalities with precise anatomical alterations in joints and soft tissues (SignalPET, 2022). Another example is Satya Nadella, CEO of Microsoft, who has made significant advancements beyond static image evaluation.

A critical application of AI lies in the screening of bovine tuberculosis (bTB), where CNNs automate the detection of granulomatous lesions in thoracic radiographs, reducing interpretation time by approximately 40 % while maintaining diagnostic accuracy comparable to expert radiologists (Rachapalli et al., 2020). Similarly, in small animal practice, AI tools assist in the critical anatomical task of differentiating malignant from benign pulmonary nodules on thoracic radiographs, thereby minimizing the need for invasive diagnostic procedures (Hespel et al., 2019). However, the efficacy of these models is intrinsically linked to the quality and diversity of the data they are trained on. A primary challenge is the need for large, well-annotated datasets encompassing a wide range

of breeds, species, and anatomical variations to mitigate algorithmic bias and ensure robust generalization (Choudhary et al., 2025). As these tools become more accessible, they promise to democratize expert-level anatomical analysis, empowering even resource-limited clinics with diagnostic capabilities once reserved for specialized referral centers (Table 1).

2.2. Predictive analytics

Predictive analytics is changing the way veterinary care is done, helping doctors anticipate health problems before they show up as symptoms. This approach goes beyond just treating what's already happening and instead focus on control of issues before they become serious. Smart computer programs use a lot of information, such as medical history and data from devices that track an animal's movement, to find early signs that something might be wrong. On dairy farms, these tools look at things like how much milk a cow is producing, how active it is, and how often it chews its cud. This helps identify issues like mastitis or metabolic diseases up to two days before the cow shows any signs of being sick (Rutten et al., 2013). This early warning allows farmers to act sooner, resulting in more effective treatment and a reduction in the use of antibiotics.

Similarly in small animal medicine, predictive models can also help to identify pets that might become ill later in life with kidney disease based on subtle changes in lab results and in weight (Gupta et al., 2024). These systems don't just use data once they keep learning as they are presented with more and more cases, and that is what makes their predictions better over time. For diseases that can jump from animals to humans, these tools rely on data from weather and animal movements and the condition of farm animals to estimate when an outbreak could occur. This gives public health authorities time to act and prevent the illness from spreading.

But there are still a few hurdles. Predictions are only as good as the data they are based on, and if a system is too sensitive, it can often raise false alarms (Gandomi and Haider, 2015). Despite these issues, predictive analytics is having a big impact continuing the revolution in health care from treating sick patients to preventing diseases before they develop. This provides veterinarians an opportunity to intervene before the illness occurs. As these tools continue to improve, they not only provide healthier pets, but they also present a more secure, proactive manner to promote global health (Fig. 1).

2.3. Decoding records: Natural language processing

Natural Language Processing (NLP) are now helping veterinary clinics extract valuable pieces of information from messy clinical notes, discharge summaries, and research papers (Cohen and Demner-Fushman, 2014). These tools can parse through the free-text entries in electronic health records and look for patterns that are difficult to discern in long, handwritten or typed notes. For example, small tips about changing behavior, or slight symptoms overlooked in a busy practice, might be snatched up automatically. It can help detect early indications of issues such as cognitive decline in older pets or pain in farm animals (Panesar and Pérez Cabello de Alba, 2023).

Complex NLP models make it easier to order and search through clinical stories. This makes it possible for people to sift through decades of medical records to identify patterns in how diseases spread or how treatments work or how drugs cause side effects. When monitoring animal-related health issues, the algorithms scour vet reports, farm records and even social media to detect fresh afflictions by identifying key words and ascertaining the overall context (Kogan et al., 2019).

2.4. AI in touching lives

AI does not just apply algorithms and sift for issues. It fosters better relationships between animals, vets and pet owners. Clinics using AI

Table 1
AI in Veterinary publications related to conventional radiography.

Species	Task	Results	Model	Reference
Orthopaedics				
dog	hip identification	sensitivity classification error	ML	(McEvoy and Amigo, 2013)
dog	classification of hip dysplasia	specificity sensitivity PPV NPV	DL	(McEvoy et al., 2021)
Thorax				
dog	detecting pulmonary coccidioidomycosis lesions	AUC	DL	(Ott et al., 2021)
dog+ human	primary thoracic lesions classifications	ROC AUC (DV radiographs) ROC AUC (LL radiographs)	DL	(Celniak et al. 2023)
dog	classification of cardiomegaly based on VHS value	AUC	DL	(Burti et al. 2020)
dog	canine MMVD stages classification	AUC for different stages	DL	(Valente et al. 2023)
Image quality analysis				
dog +cat	assessing proper collimation	accuracy	DL	(Tahghighi et al. 2023)
dog	most common technical errors determination	accuracy (LL radiographs) accuracy (sagittal radiographs)	DL	(Banzato et al. 2023)

Note: DL: deep learning; ML: machine learning; PPV: positive predictive value; NPV: negative predictive value; MMVD: myxomatous mitral valve disease; AUC: area under the curve; VHS: vertebral heart score; ROC: receiver operating characteristic; DV: dorsoventral; LL: laterolateral

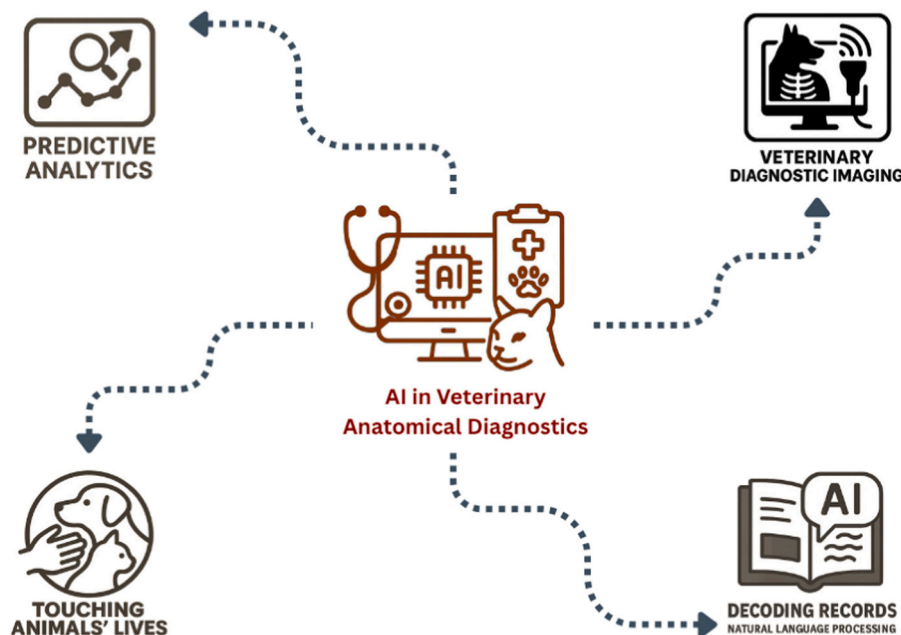


Fig. 1. AI role in Veterinary Anatomical Diagnostics.

tools do have more meaningful conversations with their clients the A.I. gets medical pictures reviewed, and that saves time (Steen et al., 2021). It provides vets with more time to speak to owners in a compassionate manner and develop individual plans for treatment. For anxious pet owners, AI inflected apps keep them updated in real time about their pet's recovery, turning complicated medical information into plain and simple visualizations. This alleviates their anxieties and gives them some assurance that their pets are being looked after.

In remote and rural areas, telemedicine tools that employ AI for diagnosing can make it easier to deliver expert veterinary care where it is otherwise hard to come by (Kogan et al. (2021). Wise farmers slap smart collars on their animals to monitor their health and receive early warnings of problems, thus preventing animals from suffering and saving money from being lost (Kaler and Green, 2013). Most significantly, AI enables vets to catch subtle signals of pain and discomfort that humans might overlook, such as changes in how animals sound or move, leading to quicker and more effective treatment.

The feeling it generates is very powerful in terms having more time is

available to be cherished pets and prevents illness that can be involve the other family members and communities at large. Technology may be getting better, but the kind, considerate job of a vet doesn't change, it can help the veterinarians in terms of additional diagnostic tools ensure that each animal gets the treatment and respect it deserves.

3. AI in zoonotic disease monitoring: Tracking the invisible

The slow spread of diseases which are transmitted from animals to humans is one of today's great challenges in medicine. But now artificial intelligence is becoming a potent new tool for identifying these hidden risks. AI includes and analyze all sorts of information like images from satellites showing where animals live and where they are moving, data from weather sensors and even posts from social media (Hassani et al., 2020). This aids in the discovery of when diseases may have made the jump from animals to people significantly faster than human monitoring systems could pick them up (Hassani et al., 2020). Such models rely on machine learning, called upon to learn from past disease outbreaks, can

uncover places in which viruses could jump from animals to humans (Carlson et al., 2022). And they are also monitoring modest changes in the landscape, such as deforestation or shifts in temperature, that might alter how diseases are transmitted (Karesh et al., 2012).

Out in the field, AI-enabled acoustic sensors track bat colonies and listen for differences in their sound that could signal an infection. Meanwhile computer vision reviews camera trap videos for any behaviors that strange in animals carrying diseases (Olumuyiwa Tolulope Ojeyinka, Toritsemogba Tosanbami Omaghomi, 2024). These tools establish an early warning system that functions far superior to outdated approaches, enabling us to discover threats like avian influenza or Nipah virus circulating months before anyone falls ill (Mehand et al., 2018). Using genomic data, AI can track how viruses evolve over time nearly in real time, capturing dangerous mutations in animal populations as they emerge.

Despite some remaining challenges, including a lack of critical information, AI's ability to link disparate data points offers the best chance to prevent the next big pandemic (Mehand et al., 2018). It also alters the way that we are watching for diseases that jump from animals to humans, and will make it possible to more easily protect both animals and people before problems start.

3.1. Early alerts

The fight against diseases carried by animals has changed with the help of AI systems that act like a global early warning system. These smart tools keep gathering and studying data from animal hospitals, wildlife monitoring locations, meat processing facilities, and health reports (Mac Loughlin, 2025). They process this information much faster than any team of people could. Strong machine learning programs find unusual patterns in how diseases spread and send alerts when strange groups of sickness appear in different places or among various animals (Saria et al., 2018).

In real situations, these systems work very well. They use smart computer programs that look at information from chicken farms in Southeast Asia and can find signs of avian flu early, sometimes within a few hours after seeing higher than usual deaths (Kamran et al., 2021). Tools that understand people's language check news and social media in different languages to find early clues about rabies or strange animal deaths that might not be reported properly (Global Public Health Intelligence Network (GPHIN) (GPHIN), 2020). By looking at weather and how animals move, these systems can guess where diseases might spread next, which helps stop outbreaks from spreading to other areas (Gostic et al., 2020).

AI analysis works fast, changing the way we track diseases by shifting from looking at past problems to stopping issues before they spread. These tools save a lot of time, reducing how long it takes to find diseases from weeks to just days, or even hours. This gives public health workers a better chance to prevent new threats from getting worse (Wang et al., 2021). It helps protect both animals and the people who are most at risk.

3.2. One health connections

Artificial intelligence is becoming the main part of the One Health approach, showing hidden links between animal, human, and environmental health in new ways. By looking at huge amounts of data from many different areas, AI finds unexpected ways diseases spread like how a virus could move from bats to farm animals and then to people, or how antibiotic resistance moves through water (Rohr et al., 2019). These systems don't just gather information; they spot the quiet connections between climate changes, animal movements, and new diseases that people might not notice.

Sophisticated network analysis shows how diseases can spread across different environments, revealing that cutting down trees in one area could lead to more cases of Lyme disease far away (Gibb et al., 2020). Machine learning uses data from both animal and human health records

to spot zoonotic infections more quickly, finding patterns that cross the usual boundaries between different medical fields (Kreuder et al., 2015). Most importantly, AI provides common language and visual tools that let vets, doctors, and ecologists work together better turning the One Health idea from just a concept into something that can actually be done (Kelly-Cirino et al., 2019; Zhang et al., 2024).

As these systems become more advanced, they have the potential to predict not only disease outbreaks but also major changes in ecosystems, helping us protect the fragile balance of life that supports all species on our connected planet.

3.3. Empowering communities

Artificial intelligence is making it easier for communities to prevent zoonotic diseases by giving them strong tools to monitor and respond. Around the world, farmers now get smartphone messages about disease risks that are tailored to their animals and where they live. Indigenous trackers also use apps powered by AI to record observations about wildlife health, and they can do this in their own languages. These tools turn local knowledge into a system that helps warn people worldwide. For example, if a village notices sick monkeys or strange bat behavior, they can share this information, and it can lead to a coordinated response across the region (Morens et al., 2020).

In areas where it's hard to get a vet, community health workers use portable AI tools that look at pictures of animal injuries or symptoms. These tools can connect with doctors who are far away when help is needed (Faust, 2018). Citizen science projects use AI to check and sort reports from people, making maps that show where diseases might be spreading in real time (Geoghegan and Holmes, 2018). Most importantly, these tools teach communities how to prevent diseases. They use simple, culturally relevant pictures to explain complicated health ideas, so no medical training is needed (Pal et al., 2019).

By connecting professional epidemiology with traditional ecological knowledge, AI helps create a new way of involving people in health monitoring where every observation count, and protecting health becomes a shared duty that's part of everyday life (Colijn et al., 2022).

3.3.1. AI and climate-driven epidemiology

Climate change plays an important role in the transmission of diseases through animals to humans due to the potential of pathogens to survive through climatic variations and infecting more number or variety of hosts etc. Artificial intelligence helps understand these complex connections by combining multiple kinds of climatic and environmental data sets into predictive models to predict the future outbreaks and their potential impacts. For example, machine learning have the potential to analyze long term changes in temperature, humidity, and extreme weather to forecast where ticks those carry Lyme disease might move, helping to predict higher disease risk in new areas (Gibb et al., 2020). Also, AI models that use data from satellites about ocean temperatures and rain patterns have been used to predict outbreaks of Rift Valley fever in animals, allowing for early vaccination efforts (Anyamba et al., 2009).

Beyond just looking at weather, AI is really good at understanding changes in the environment caused by human activities. These smart systems can look at satellite pictures to track things like trees being cut down, cities growing, and farms expanding in real time. They can predict the factors responsible for the distortion of natural animal habitats, pushing closer contacts of wildlife humans and their pets. This makes it more likely easier outbreaks of the diseases like Nipah and Ebola virus through human-animal contact (Carlson et al., 2022). These risk maps generated by AI don't work alone, they combine information about future climate changes with data regarding the populations of livestock and mobility of wildlife species, and urbanization due to intensive human populations. This also helps to generate detailed and up-to-date maps showing where diseases can integrate in future due to climatic/environmental changes. This new way of understanding risks prepares for the future risks, monitoring, efficient use of resources and their

strategic control to avoid huge economic and social impacts due to zoonosis.

4. Challenges

4.1. Anatomical variability and model generalization

A fundamental challenge in applying AI to veterinary medicine is the significant anatomical diversity among species, breeds, and individual animals, which poses serious challenges to the generalization ability of computational models, and this limitation has been identified in recent studies also. These studies point out that although artificial intelligence systems are very powerful, however, they often struggle with the "spectrum of normality" in animal anatomy. For example, a concept that experienced clinicians intuitively understand but is hard to translate into algorithms (Choudhary et al., 2025). This spectrum is not just theoretical; it has direct clinical implications. Similarly, AI-powered techniques can produce errors when applied to patients whose anatomical features fall outside the model's original training data (Vickram et al., 2025).

The issue gets even worse because many AI models are created and tested using data from animal models that are set up for research. These models don't show the different body structures found in real patients (Choudhary, 2025). For instance, a deep learning model trained to find broken bones in a lab dog with a simple body shape might not work well for a Basset Hound with a very sharp angle in its bones or a sugar glider, having different skeletal presentation.

In addition to gathering data, the solution also involves literacy of the end users about these models, for example as future veterinarians start learning anatomy with tools like ChatGPT, it's important that these tools are trained on a wide range of accurate anatomical information. This will help to avoid any unfair assumptions and make sure new veterinarians understand the many different ways body structures can vary (Choudhary et al., 2023). Overcoming the challenge of different body structures isn't just about fixing technology, it needs everyone around the world to work together to create detailed and properly labeled image collections. These collections should show the full variety of animal body shapes, so AI can be a dependable tool for all patients, not just for few.

4.2. Data gaps

The promise of AI in veterinary medicine faces a serious challenge like there's not enough good data to make it work well. In human medicine, there are big collections of standard data that help AI learn. But in veterinary care, data is scattered because clinics use different systems that don't talk to each other. Also, many records are written by hand and are hard to turn into digital information (Hernandez et al., 2022). There's also a big problem with how different animals are represented in the data there's way more information on dogs than on exotic animals (Kogan et al., 2021). Plus, diseases in livestock in poorer countries are not reported enough, even though they affect people around the world.

Even when data is available, it often misses important details like how treatments worked, follow-up care, or environmental influences that could show important trends. Special veterinary systems make it hard to share information between places, and worries about privacy stop hospitals from sharing patient data (Jameson and Longo, 2015). This leads to AI tools that are built using limited and not very representative data. These tools might work well in studies, but they don't perform as well in actual clinics where patients are more varied (Topol, 2019).

One of the biggest concerns is the "silent epidemic" issue: diseases that aren't found because there's no way to test for them in certain areas, and so there's no record for AI to study (Branswell, 2017). Unless we fix these basic problems with data by creating common ways to report

information, sharing data freely, and making sure everyone is included in collecting data, veterinary AI might end up helping only some people while leaving others without support.

4.3. Trusting the unknown

The use of AI in veterinary work runs into a big human problem, for example people are unsure if they can trust these AI systems to make important decisions about life and death (Castelvecchi, 2016). Many vets are hesitant to follow AI results that don't match what they've learned through years of training and their own sense of what's right, especially when the AI can't explain why it came up with that result in medical terms (Rajkomar et al., 2018). This issue of not understanding how AI works becomes even bigger in difficult cases where small signs or details from the patient's history might not be picked up by the system (Ribeiro et al., 2016).

Practice managers face various trust challenges, such as doubts about how reliable the system is, worries about being held responsible if AI makes mistakes, and questions about whether the technology is worth the cost (Ferryman and Pitcan, 2024). Pet owners also bring added complexity, with some people not liking AI-based diagnoses because they feel they lack a personal touch, and others doubt veterinarians who don't agree with the recommendations from online algorithms (Binns et al., 2018).

The trust gap shows up in clinics where staff still keep records by hand as a backup, even though they have AI systems. This creates extra work. To fix this, it's not just about making the AI more accurate. It needs to act like a team member, not just a smart tool. It should explain its reasoning clearly, admit when it's unsure, and learn from the doctors' input (Holzinger et al., 2017). Until then, even the best AI will still face the human need to understand before trusting (Grote and Berens, 2020).

4.4. Ethical questions

The fast use of AI in animal healthcare is causing big ethical questions that the field is just starting to deal with (Adams, 2021). When does using AI to help make diagnoses go too far? Should things like how much a pet owner can afford to pay affect what AI suggests for treatment? These issues get even trickier when looking at animal medicine for farms, where AI choices might focus more on making the whole group of animals more productive than on taking care of each animal's well-being (Thompson, 2020).

Data privacy is a major worry as farmers are right to be concerned about who has control over the health information of their animals and how that data might affect their insurance costs or their ability to sell their products (Lutz and Hoffmann, 2021). At the same time, using a lot of energy for AI tools is also a problem, because even though these systems are meant to help animals, they also contribute to climate change, which harms all living things (Strubell et al., 2019).

The significant worry is that AI systems might have unknown unfairness. That happens when the system is trained on data from well-appointed clinics that aren't seeing the health problems that plague stray dogs, or working dogs (Veale and Binns, 2017). As AI becomes more prevalent in making medical decisions, the veterinary industry must develop guidelines on ethics. To prevent technology outpacing the virtues of judgment and care that are crucial to better animal health care (Vayena et al., 2018).

4.5. Bridging the divide

The idea of using AI in animal healthcare isn't available to everyone, which makes the difference between top clinics and smaller ones even bigger. Specialized hospitals use smart tools to find out what's wrong with animals, but vets in rural areas often can't use these tools because they have slow internet that doesn't work well with AI systems that rely on the cloud (Wightman et al., 2022). This technology gap shows there

are other bigger issues too like places that don't have good animal care also often don't have good human healthcare or good education (Grace et al., 2012).

New methods are being created to deal with this problem. Small AI models that don't need an internet connection can run on mobile phones and provide powerful diagnostic tools to places where there isn't reliable electricity. Veterinary schools are now including AI in their main classes, so new doctors can use these tools wherever they practice (Nyamuryekung'e et al., 2023). A major change is the use of shared platforms where doctors worldwide share AI tools designed for their specific local needs such as mobile apps that can identify animal diseases with just a small amount of information.

Real progress in AI for veterinary care isn't just about sharing technology but it needs systems that can work in different places, not just in perfect hospital settings. This means creating tools that can be used in many languages, using data from real working animals and small farms, and pricing that works for everyone, including smaller clinics. Only by making sure everyone has access from the start can AI really help improve animal health all around the world (Ostfeld et al., 2008).

5. Future directions

5.1. Creating robust data

The way to create trustworthy veterinary artificial intelligence starts by tackling the data problem. It's not just about gathering more data, but about getting better, more accurate data that represents all types of animals. New methods are being developed to share data between different places, like using secure platforms based on blockchain. These let clinics send anonymized patient records while keeping patient privacy (Ainsworth et al., 2020). Teams from different organizations are also working together to build databases focused on specific animals, making sure to include those that are often left out, like exotic pets and working horses (Batchelor et al., 2022).

New ways of collecting data are turning old limits into new chances. When farmers record video diaries of animal behavior and use AI to analyze them, it creates detailed information about how animals act, without needing costly sensors (Cuzzolino, Fernandez Pierna, 2021). Systems called federated learning let AI improve using data spread across different places, without moving the data from where it's collected, which helps protect privacy and makes the AI better (Kaissis et al., 2020). Most importantly, worldwide efforts are now setting standard formats for animal health data, so AI can learn from a wide variety of animals instead of just a small group.

The next big step is about creating "data bridges" that link veterinary records with environmental sensors, wildlife health tracking systems, and human health databases (Weitzman et al., 2019). This helps build a full information system that's essential for really powerful AI uses. It's not just about giving data to machines; it's about creating a strong knowledge base that will improve veterinary medicine for many years ahead (Veldhuis et al., 2023).

5.2. Opening the black box

The veterinary field needs more than just correct AI predictions, it also needs clear explanations. New explainable AI (XAI) methods are changing how complex algorithms work by making their decisions easier to understand. These tools help show how AI looks at X-rays, pointing out exactly where bones are broken or where soft tissues might be affected. Clinicians can now use interactive dashboards to try different options by changing the input settings and instantly see how the AI's results change (Tjoa and Guan, 2020).

For difficult cases, natural language generation turns algorithm results into easy-to-read reports that show how a vet would think: "The moderate subpleural opacity in the left caudal lung lobe, along with the patient's ongoing cough, points to early-stage pneumonia with 78 %

confidence" (Ghassemi et al., 2021). Some systems also share how sure they are about their findings, which is important when the situation isn't clear.

The best platforms use feedback from veterinarians, where their input and comments help improve the AI's decisions over time (Topol, 2019). This teamwork helps build trust and makes the AI more accurate. It doesn't replace the vet's knowledge, but instead creates a conversation between human judgment and computer analysis. As these tools get better, they aim to make AI not just a helper, but someone that can learn from vets, supporting the skills and care that are central to good veterinary medicine (Gilpin et al., 2018).

5.3. Designing ethical AI

Today, as AI for veterinary medicine is being developed, the emphasis is not just on how well the technology works, but on ensuring it's ethical every step of the way. The initiative takes a more expansive view of AI than a narrow technical one that considers how artificial intelligence affects society at large. Professionals in groups like veterinarians, ethicists and animal welfare specialists are creating rules for using AI in a responsible manner (Borenstein et al., 2017). These guidelines also contain regular reviews to identify and address any unfairness in AI systems, to ensure that diagnostic tools perform well and treat all animals in a fair way, regardless of their breed, type, or origin (Whittlestone et al., 2019).

New ways of applying AI in veterinary care also ensure that veterinarians get the final say on AI recommendations, so that the algorithms act as advisers rather than making decisions on their own (Gerke et al., 2020). These programs also adhere to privacy-by-design approaches to keep patient data safe from the start and allow clinics to work with AI applications without losing control of essential medical documents (Shaban-Nejad et al., 2018). Another new idea is the use of "compassion metrics" by some platforms to assess treatment options not just on how effective they are, but on how much they decrease pain and improve the pet's quality of life (Sarkar and Chen, 2021).

It's also just about thinking about environmental ethics. Developers are making algorithms run with less energy, and choosing cloud servers that use renewable energy (Kaack et al., 2022). Those are the steps you see here that demonstrate what veterinary AI should and should not be based on the values of the profession. It's not solely about obtaining accurate results, but also about employing technology that is respectful of animals and assures clients, and that supports the environment. This, in turn, is AI supporting the best values we have in veterinary medicine, rather than gaming them for the path of least resistance (Mittelstadt and Floridi, 2016).

5.4. Making AI accessible

And the full potential of veterinary AI can only be achieved when it's provided everywhere, in every clinic, farm, zoo, and other place where animals are cared for, regardless of where or what resources they have. It's developers who are breaking down the barriers, building simple algorithms that can be turned into powerful diagnostic tools on inexpensive smartphones and tablets (Srivastava et al., 2021).

What would represent the true power of AI in the veterinary field would be for it to be available in every clinic, indeed on every farm, and in whatever spaces that animals are kept wherever they are, and by whatever means. Software developers are cracking the code with simple algorithms that offer powerful diagnostic tools on inexpensive smartphones and tablets. The basic versions are still very good and are made to handle real world issues such as slow internet connections or older devices (Bansal et al., 2020).

The best solutions are ones that result from close collaboration with local partners, as when developers pair up with veterinarians in various locales to create tools that fit the local community's specific needs (Poupaud et al., 2021). And this work ensures AI isn't something that

comes from big, well-heeled places but grows naturally to aid all animals and the people who care for them, democratizing what was once a niche technology into a broadly available standard of care.

6. Conclusion

As we think about how AI is changing veterinary medicine and how we watch for diseases that can jump from animals to humans, one thing is clear: technology's main goal isn't to take the place of our bond with animals, but to make that connection stronger by showing us a new way to understand their bodies. The algorithms we've talked about aren't just cold, unfeeling computers; they're smart tools that, when used well, can help animals that can't tell us when they're hurt by finding small health problems that humans can't always see. They help busy vets by doing the detailed work of analyzing complex body structures automatically, and they act as smart defenses against diseases that could spread around the world.

The path to this future is marked by both hope and caution. The successful implementation of AI depends on our capacity to address the fundamental issue of anatomical variation, ensuring that models are trained on data representing the full range of animal diversity, as highlighted in studies exploring AI's potential and limitations in anatomy. The AI techniques we embrace must enhance, rather than diminish, the clinician's understanding, acting as a collaborative partner in diagnosis and surgical planning. Another example is the foundation of trust established through explainable systems and ethical guidelines that prioritize animal welfare and equitable access.

Ultimately, the future we need to create is one where AI takes care of the things it does best like recognizing patterns in big sets of data and doing detailed analysis of anatomy. This way, veterinarians can focus on what they do best: helping animals heal with both skill and compassion. Whether in a research lab using strong animal models, a small clinic in a rural area, or a worldwide system that watches for diseases, we have a chance and a duty to make technologies that help all animals, big and small. If we do this right, based on the One Health approach, we won't just have smarter medicine. We'll have a world where technology helps us take better care of animals, react quicker to problems, and protect with more kindness than ever before. This will help create a healthier future for animals, people, and the planet we all share.

Declaration of Competing Interest

It is declared that authors have no competing interests or personal relationships that could have appeared to influence the work reported in this paper.

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